Fiscal Year:	FY 2010	**Task Last Updated:**	FY 12/22/2009
PI Name:	Goree, John Ph.D.		
Project Title:	Self-Structuring in Dusty Plasmas (NNX07AD22G)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received funding continuation to 4/14/2013, per NSSC (8/2010) NOTE: Received extension to 2/28/11 (4/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Morfill, Gregor (Max Planck Institute for Extraterrestrial Physics)		
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Task Description:

This fundamental research is an experimental study of dusty plasma physics. Data are analyzed from a European instrument, PK-3 Plus, presently on orbit on the International Space Station ISS. (The acronym PK refers to "Plasma Crystal," and ISS is the International Space Station.) Ground-based laboratory experiments are carried out to prepare for future microgravity flight experiments, including the planned PK-4 instrument.
Dusty plasma consists of particles of solid matter suspended in ionized gas. Micron-size particles acquire a large negative charge of typically 10,000 elementary charges. Due to inter-particle repulsion, they arrange themselves in structure with solid or liquid-like properties, similar to a colloidal crystal. With particles of this size, gravity is a dominant force, causing particles to sediment to a thin layer at the bottom of the plasma.

Under microgravity conditions, particles fill a 3D volume and the suspension self-organizes. We study: • Structural features. These include the boundaries of the particle suspension, including a void at the plasma's center and vortices at its edges. • Dynamical behavior exhibited by the dusty plasma. These include waves, instabilities, and transport. This dynamical behavior is studied at an atomistic level, using video micrography to observe the motion of microspheres which serve as an analog for atoms in normal molecular matter.

Experimental methods include direct imaging of particle suspensions using video micrography and particle tracking, and manipulation of particles using the radiation-pressure force of powerful cw lasers.

Ground-based laboratory experiments accompany the analysis of data returned from space. The ground-based work serves the dual purpose of developing scientific methods to be used in future flight experiments as well as providing an immediate scientific return.

The spaceflight microgravity experiments are performed using European instruments on ISS: PK-3 Plus (presently on orbit) and PK-4 (planned). NASA is not expected to provide any flight hardware. Video cameras are used to image not only the particle suspension as in previous experiments, but also spectral emission from the plasma glow. These space-flight experiments on ISS are carried out under the direction of German and Russian scientists. The PI is a team member on both the PK-3 Plus and PK-4 missions, and receives the data after it is returned from the ISS.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Status of ISS missions:
During this period, the PI, Professor Goree, was a team member for two missions on the International Space Station (ISS): The PK-3 Plus mission is primarily a German-Russian collaboration, and the instrument is presently on-orbit on the ISS and returning data. The PK-4 mission is primarily a European Space Agency (ESA) mission, and is now in the planning stages. The acronym PK refers to the German spelling of "Plasma Crystal." Both projects concern dusty plasma, also known as complex plasma, which consists of micron-size particles of solid matter immersed in ionized gas.

Analysis of ISS data:

During this period, Professor Goree received two shipments of data from PK-3 Plus. Data that were returned aboard Soyuz spacecraft were delivered on computer hard disks to Professor Goree in Iowa. The data consist of digitally-recorded video accompanied by time series recordings of voltage, temperature, and other housekeeping data. After receiving these shipments, the PI now has five hard disks of data from PK-3 Plus.

Data analysis of PK-3 Plus data has been carried out, and remains underway. This work is being done by two members of Professor Goree's group.

The main scientific topic being explored at Iowa using PK-3 Plus data concerns transverse waves in the suspensions of microparticles. Our first effort focuses on waves in a single-layer suspension of microparticles, and our second effort on three-layer suspensions. These waves have a spectrum of frequencies and wavenumbers, corresponding to natural random thermal motion. By analyzing images to track the motion of particles, we are computing the experimentally-observed spectrum using the PK-3 Plus data. We are comparing these results to theoretical dispersion relations. The theory is for sound waves in a monolayer suspension of particles that interact through a shielded Coulomb repulsion.

Our first publication based on PK-3 Plus data appeared in 2009 in Physics of Plasmas. The title was Transverse Oscillations in a Single-Layer Dusty Plasma under Microgravity. Russian and German team members were included as co-authors.

A second publication based on PK-3 Plus data is now in the final stages of preparation. The topic is the development of a new method of measuring charge and other in-situ parameters in microgravity dusty plasma experiments, based on video imaging of random particle motion.

Ground-based laboratory research:

During this period, Professor Goree's group also carried out ground-based laboratory research to support the flight experiments. Microgravity enables the study of three-dimensional suspensions of microparticles, unlike laboratory conditions where the particles usually sediment to form a thin horizontal layer.

The laboratory experiments underway focus on a type of longitudinal wave called the "dust density wave." These type of waves are often observed in PK-3 Plus. To prepare for future analysis of PK-3 Plus data, we are carrying out laboratory experiments to observe similar waves under gravity conditions. These laboratory tests help identify the experimental methods and data analysis methods that we will be able to use in future experiments using PK-3 Plus. The experiments center on identifying the mechanisms that provide energy to the waves, and conversely mechanisms that damp the wave.

We published a laboratory-based paper in 2009 on a gas-flow topic that will help in the design and interpretation of future microgravity experiments. The title of this paper was "Thermal Creep Flow Detected in Dusty Plasma," and it reported the discovery that the kind of lasers used for manipulation can cause unwanted gas flows when they pass through certain types of glass windows.

	We submitted a laboratory-based paper in 2009 on a fundamental research problem that will influence our future experiments on PK-4. In this experiment, we used laser manipulation to suddenly apply shear to a crystalline suspension, causing it to suddenly melt. We investigated the spatio-temporal development of the melting front. This experiment was done with a single-layer (two-dimensional) suspension, as compared to the three-dimensional suspension that will be used in future PK-4 experiments. The title of this paper was “Evolution of shear-induced melting in dusty plasma.” Computer simulation research: Computer simulations of the experiments are an important part of analyzing experimental data as well as preparing for future experiments. During this period, Professor Goree's team collaborated with a leading simulation group in Budapest Hungary, led by Dr. Zoltan Donkó. The simulations tested whether transport coefficients, such as diffusion, viscosity and thermal conductivity, are meaningful for an analog liquid with shielded Coulomb interparticle interactions. This resulted in a paper that was published in 2009 with the title: “Time Correlation Functions and Transport Coefficients of Two-Dimensional Yukawa Liquids.” Planning of future space missions: During 2009, the PI attended ESA’s meeting of the PK-4 Facility Science Team, held in Garching Germany. The PI was elected Vice-Chairman of the team. During this meeting, input from scientists was collected to guide ESA during the engineering Phase C/D for this mission.
Bibliography Type:	Description: (Last Updated: 01/23/2025)
Articles in Peer-reviewed Journals	Donkó Z, Goree J, Hartmann P, Liu B. "Time-correlation functions and transport coefficients of two-dimensional Yukawa liquids." Physical Review E, 2009 Feb;79(2 Pt 2):026401. PubMed PMID: 19391849 , Feb-2009
Articles in Peer-reviewed Journals	Liu B, Goree J, Fortov VE, Lipaev AM, Molotkov VI, Petrov OF, Morfill GE, Thomas HM, Rothermel H, Ivlev AV. "Transverse oscillations in a single-layer dusty plasma under microgravity." Physics of Plasmas, 2009 Aug;16(8):083703. http://dx.doi.org/10.1063/1.3204638 , Aug-2009
Articles in Peer-reviewed Journals	Flanagan TM, Goree J. "Gas flow driven by thermal creep in dusty plasma." Physical Review E, 2009 Oct;80(4 Pt 2):046402. PubMed PMID: 19905456 , Oct-2009
Articles in Peer-reviewed Journals	Arp O, Caliebe D, Menzel KO, Piel A, Goree JA. "Experimental Investigation of Dust Density Waves and Plasma Glow." IEEE Transactions on Plasma Science, Accepted as of December 2009. Expected publication February 2010. , Dec-2009